Smart Fleet Management Database

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Background

A logistics company operates a fleet of delivery vans, refrigerated trucks and service vehicles across several cities in Vietnam. Vehicles are used for parcel delivery, equipment transport and field servicing.

The fleet currently operates from depots in:

- Ha Noi
- Da Nang
- Ho Chi Minh City
- Can Tho

Each depot manages a group of vehicles and drivers. Each depot also contains workshop facilities where maintenance and repairs are performed.

The company has expanded rapidly over the last few years. Information is currently stored across spreadsheets, workshop whiteboards and several disconnected software systems. Managers have reported difficulty tracking:

- Vehicle availability
- Driver assignments
- Safety incidents
- Workshop activity
- Maintenance history
- Supplier pricing
- Vehicle downtime

The company recently installed telematics devices and onboard diagnostic systems in all vehicles. These systems generate information about driver behaviour and vehicle condition.

The raw telemetry itself is processed by external systems, however important operational events and alerts still need to be stored in the company database.

The company wants a relational database system capable of supporting fleet operations, driver monitoring and workshop management.

Vehicle information includes:

- Registration number
- Vehicle identification number (VIN)
- Vehicle category
- Manufacturer and model
- Year of manufacture
- Current odometer reading
- Depot location
- Operational status

Vehicles may be classified as:

- Delivery Van
- Refrigerated Truck
- Electric Van
- Service Vehicle
- Heavy Transport Truck

A vehicle may be:

- Active
- Available
- Under Maintenance
- Awaiting Inspection
- Out of Service
- Retired

Some vehicles are permanently assigned to a driver, while others are shared between multiple drivers depending on operational requirements.

Vehicle assignments change regularly between depots and drivers.

A vehicle that is currently under maintenance or marked out of service cannot be assigned to a driver.

The company maintains a complete maintenance history for all vehicles.

Drivers and Certifications

Drivers employed by the company may operate different vehicle categories depending on their licences and certifications.

Driver information includes:

- Driver identification number
- Full name
- Contact information
- Depot assignment
- Licence type
- Licence expiry date
- Employment status
- Emergency contact details

The company tracks driver certifications because some vehicle categories require specialised qualifications.

Drivers cannot be assigned to vehicles unless they hold the required certifications for that vehicle category.

Some certifications expire and require renewal.A driver with expired certificates cannot be assigned to a vehicle.

Vehicle Certification Matrix

To drive the following vehicles you need ALL the certifications as follows:

Vehicle Type	Standard Licence	Heavy Vehicle Licence	Refrigerated Transport Certification	EV Certification	Hazardous Goods Certification
Delivery Van	X				
Refrigerated Truck	X	X	X		
Electric Van	X			X	
Service Vehicle	X				
Heavy Transport Truck		X			X

The company has identified that some drivers hold multiple certifications. Drivers may also renew or upgrade certifications over time.

Example driver certifications currently recorded by the company include:

Driver ID	Driver Name	Certification	Expiration
D-112	Nguyen Van An	Standard Licence	14/02/2025
D-112	Nguyen Van An	EV Certification	30/04/2025
D-204	Tran Thi Bich	Heavy Vehicle Licence	8/12/2025
D-204	Tran Thi Bich	Refrigerated Transport Certification	1/5/2026
D-331	Le Quoc Minh	Standard Licence	21/06/2026
D-417	Pham Duc Long	Hazardous Goods Certification	18/11/2026

Driver Behaviour Monitoring

Each vehicle contains a telematics device that records driver behaviour events. The company stores important events generated by the telematics system.

Examples include:

- Harsh braking
- Rapid acceleration
- Excessive speeding
- Sharp cornering
- Excessive idling
- Fatigue warnings
- Seatbelt violations
- Phone distraction alerts

Events are associated with:

- A driver
- A vehicle
- A depot
- A timestamp
- A severity level

Severity levels include:

- Low
- Medium
- High
- Critical

Some incidents automatically trigger review by fleet safety staff. High or Critical events will automatically trigger a review.

Operations staff may add comments or recommendations after reviewing incidents.

Drivers with repeated high-severity incidents may be required to attend additional safety coaching or retraining. If a critical event happens the driver will be made inactive and unable to be assigned to a vehicle until the review has been completed or he completes the safety training.

The company calculates monthly driver safety scores based on recorded events. A driver has a score of 100 each month. This score is then reduced by the following event penalties.

A driver with a score of 75 or below must attend driver coaching. A driver with a safety score of 50 or below cannot be assigned to a vehicle until they complete safety training.

Event Penalties

Event Severity	Points Deducted
Low	-2
Medium	-5
High	-10
Critical	-20

Condition	Additional Deduction
More than 3 speeding events in a month	-10
More than 2 fatigue warnings in a month	-15
Any critical event	-10

Fleet managers often compare safety performance between depots and driver groups.

Example Driver Safety Event Log

Before introducing a centralised database system, driver incidents were recorded using spreadsheets and paper forms.

EventID	Timestamp	VIN	Vehicle Type	Depot	DriverID	Driver Name	Event Type	Severity	Odometer
E091	2024-05-10 08:14	29A-123.45	Delivery Van	Ha Noi	D-112	Nguyen Van An	Harsh Braking	Low	45,100
E092	2024-05-10 09:30	29A-123.45	Delivery Van	Ha Noi	D-112	Nguyen Van An	Speeding	High	45,140
E093	2024-05-11 11:00	51C-789.01	Refrigerated Truck	HCMC	D-204	Tran Thi Bich	Sharp Cornering	Medium	112,050
E094	2024-05-12 14:20	43E-456.78	Electric Van	Da Nang	D-112	Nguyen Van An	Fatigue Warning	High	12,300
E095	2024-05-13 07:42	29A-123.45	Delivery Van	Ha Noi	D-331	Le Quoc Minh	Excessive Idling	Low	45,310
E096	2024-05-13 18:05	51C-789.01	Refrigerated Truck	HCMC	D-204	Tran Thi Bich	Speeding	Critical	112,480

Fleet managers report that information is often duplicated across spreadsheets, resulting in inconsistent vehicle names, depot names and severity labels.

Vehicle Maintenance

The onboard diagnostic systems generate predictive alerts when abnormal behaviour is detected in vehicle systems. Examples include:

- Brake wear warnings
- Engine overheating risk
- Battery degradation
- Oil quality deterioration
- Transmission fault warnings
- Cooling system anomalies
- Tyre pressure irregularities

When an alert is generated, workshop staff may take one of the following actions:

- Acknowledge the alert and continue monitoring the vehicle
- Schedule a future inspection or service
- Escalate the alert for urgent repair
- Mark the alert as resolved without further action

An alert does not always result in a maintenance job. However, when a job is created in response to an alert, the alert must be linked to that job so the outcome can be tracked. A maintenance job may also be created independently of any alert, for example, as part of a scheduled service interval.

Each workshop visit is recorded as a maintenance job. A job is the central record linking a vehicle, a workshop, a set of mechanics, and all work performed during that visit.

A job may consist of one or more maintenance activities. Each activity represents a distinct type of work performed during the job, such as a brake service or a diagnostic test.

Job-Level Information

The following information is recorded for each job:

- Vehicle
- Workshop location
- Job open date and close date
- Overall vehicle downtime
- Total maintenance cost
- Linked predictive alert, if applicable

This is what a maintenance job record looks like in the company at the moment

Job ID	VIN	Vehicle Type	Depot	Workshop	Date opened	Date closed	Down time (hrs)	Total cost (VND)
M1021	*29A-123.45*	*Delivery Van*	*Ha Noi*	*Ha Noi Central*	*2024-05-12 09:00:00*	*2024-05-13 03:00:00*	*18*	*1,800,000*
M1022	*51C-789.01*	*Refrigerated Truck*	*HCMC*	*HCMC South*	*2024-05-14 08:00:00*	*2024-05-14 14:00:00*	*6*	*3,610,000*

Activity-Level Information

The following information is recorded for each activity within a job:

- Activity type
- Mechanics assigned to the activity
- Labour hours per mechanic
- Diagnostic result
- Repeat fault indicator (whether the same fault has occurred before on this vehicle)
- Warranty indicator (whether a warranty claim applies to this activity)

This is what an activity record looks like in the company at the moment

Job ID	Activity ID	Activity type	Mechanic ID	Mechanic name	Labour hours	Diagnostic result	Repeat fault?	Warranty?
M1021	*1*	*Brake Service*	*ME-12*	*Hoang Van Duc*	*2.5*	*Pads worn below 3mm*	*N*	*N*
M1021	*1*	*Brake Service*	*ME-15*	*Pham Thi Lan*	*2.5*			
M1021	*2*	*Tyre Replacement*	*ME-12*	*Hoang Van Duc*	*1.0*	*Worn unevenly - possible alignment issue*	*Y*	*N*
M1022	*1*	*Preventative Servicing*	*ME-07*	*Nguyen Thi Mai*	*1.5*	*OK*	*N*	*N*
M1022	*2*	*Refrigeration Repair*	*ME-09*	*Tran Quoc Bao*	*2.0*	*Belt cracked - 3rd replacement this year*	*Y*	*Y*

Workshops and Mechanics

The company operates one workshop per depot. Each workshop contains multiple service bays and employs mechanics with different certifications and specialisations.

Some activities require several mechanics working together. Mechanics may be assigned to multiple activities across different jobs. Certain activity types require mechanics who hold a specific certification. A mechanic cannot be assigned to an activity unless they hold the required certification. The table below defines the certifications required:

Activity Type	Required Mechanic Certification
Routine Inspection	Standard Vehicle Mechanic Licence
Preventative Servicing	Standard Vehicle Mechanic Licence
Diagnostic Testing	Standard Vehicle Mechanic Licence
Emergency Repair	Standard Vehicle Mechanic Licence
Component Replacement	Standard Vehicle Mechanic Licence
EV Battery / Electrical Repair	EV Technician Certification
Refrigeration System Repair	Refrigeration Systems Certification
Heavy Vehicle Repair	Heavy Vehicle Mechanic Licence

Extension Tasks

Students who complete the core schema and queries are encouraged to tackle the following extension:

1. Parts used during each activity: each activity may consume multiple parts, and each part has a part number, description, unit price, quantity used, etc.
2. Supplier management: each part has a designated primary supplier and an optional backup supplier, each with their own unit cost. Supplier records include contact information and delivery lead times.
3. Warranty claims linked to specific parts: a warranty claim may cover one or more parts replaced during an activity. The database must record whether the warranty is with the vehicle manufacturer or the parts supplier.
4. Mechanic profiles and certification renewal: model mechanics as full entities with employment details and assigned workshop. Certifications must be tracked with issue and expiry dates, and the full renewal history retained so past job assignments can be verified against qualifications held at the time.

Use of Fleet Database

There are two major stakeholders in the system:

1. Fleet safety operations staff responsible for driver behaviour monitoring.
2. Workshop management staff responsible for maintenance and repair operations.

Fleet Safety Operations Staff

Fleet safety staff need to:

- Review driver incidents
- Monitor high-risk drivers
- Compare safety trends between depots
- Track licence expiry dates
- Monitor unresolved incidents
- Record coaching outcomes
- Identify drivers requiring retraining

Drivers may generate many incidents over long periods of time, so managers need to search and filter records by:

- Driver
- Vehicle
- Depot
- Event type
- Severity
- Date range

Drivers need to be able to view their own safety history and compare monthly safety scores over time.

The company also wants to identify:

- Drivers with repeated speeding incidents
- Vehicles associated with severe incidents
- Drivers with expired certifications
- Drivers operating outside their authorised vehicle categories

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Workshop Management Staff

Workshop managers need to:

- Monitor predictive maintenance alerts
- Identify vehicles requiring urgent repair
- Track workshop workload
- Allocate mechanics to jobs
- Record parts usage
- Monitor supplier performance
- Review vehicle downtime
- Compare maintenance costs between vehicle models

Managers also need to identify:

- Vehicles overdue for service
- Vehicles with repeated component failures
- Parts below reorder thresholds
- Vehicles awaiting inspection
- Mechanics with required certifications

Mechanics need access to:

- Vehicle maintenance history
- Diagnostic records
- Previous repair information

Historical Records

Historical records must remain available even when:

- Drivers transfer depots
- Certifications expire
- Suppliers change
- Vehicles are reassigned
- Maintenance rules are updated
- Predictive alert thresholds are adjusted